Quiz1 : solns.

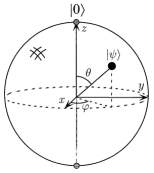
1. (i) Write the Hamilton's equations of motion in classical mechanics for generalised coordinates $\vec{q}$ and $\vec{p}$.
Ans. $\vec{\nabla}_{\vec{q}} H = -\dot{\vec{p}}$, $\vec{\nabla}_{\vec{p}} H = \dot{\vec{q}}$

(ii) What does the double slit experiment (both slits open) with electrons demonstrate?
Ans. The interference pattern obtained on the screen similar to light shows that electrons have wave like properties.

(iii) Which of the following is(are) NOT appropriate wavefunction(s) for a natural system?
(a) e^{-x}; (b) e^{-x^2}; (c) e^{-x^4}; (d) $\cos x.e^{-x^2}$ (e) $\cos^2 x.e^{-x^2}$
Ans. (a) e^{-x}; Expl. The function diverges for $x \to -\infty$, so it cannot be an appropriate wavefunction for a natural system.

(iv) How is the canonical basis vector $|0\rangle$ represented in Bloch sphere?

Ans.



As a radial unit vector in positive z-direction

Expl. : Bloch sphere representation: $|\psi\rangle = \cos\frac{\theta}{2}|0\rangle + e^{i\phi}\sin\frac{\theta}{2}|1\rangle$. For vector $|0\rangle$, $\cos\frac{\theta}{2} = 1$ and $\sin\frac{\theta}{2} = 0 \implies \theta = 0$.

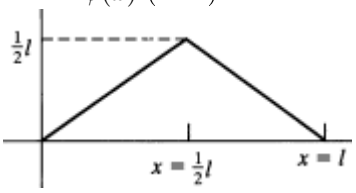
(v) Is $\frac{d}{dx}$ a Hermitian operator?
Ans. $\because$ momentum, $\hat{p}_x = \frac{\hbar}{i}\frac{d}{dx}$ is Hermitian, $(\hat{p}_x)^\dagger = \hat{p}_x \ \therefore \left(\frac{d}{dx}\right)^\dagger = -\frac{d}{dx}$ cannot be Hermitian.
One can also see this from $\int \psi_1^* \frac{d}{dx}\psi_2 dx = -\int \left(\frac{d\psi_1}{dx}\right)^* .\psi_2 dx$, or,$\langle\psi_1|\frac{d}{dx}|\psi_2\rangle \neq \langle\frac{d}{dx}\psi_1|\psi_2\rangle$

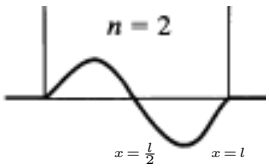
(vi) Simplify the commutator $\left[\frac{\hat{p}_y}{\hbar},\ \frac{\hat{y}}{\hbar}\right]$ is
Ans. (c) $\frac{1}{i\hbar}\times$ identity operator; Expl. : $[\hat{p}_y,\ \hat{y}]\, f(y) = \left[\frac{\hbar}{i}\frac{\partial}{\partial y}y - \frac{\hbar}{i}y\frac{\partial}{\partial y}\right] f(y) = \frac{\hbar}{i}\frac{\partial}{\partial y}(yf) - \frac{\hbar}{i}y\frac{\partial f}{\partial y} = -i\hbar f(y) \implies \left[\frac{\hat{p}_y}{\hbar},\ \frac{\hat{y}}{\hbar}\right] = -\frac{i}{\hbar}\hat{1}$

2. (i) What are the dimensions of a 1-D wavefunction?
Ans. (a) dimensionless; (b) $L^{-\frac{1}{2}}$; (c) L;(d) that of energy
details : $\because \int |\psi|^2 dx$ is a probability (dimensionless), $\therefore$units of a 1-D wavefunction are $L^{-\frac{1}{2}}$

(ii) If $|n\rangle$ denotes an eigenstate for the particle in a 1-D box problem, and given **:** $\psi(x) = x(l-x)\,\forall\, 0 < x < l$, obtain the value of $\langle n = 2|\psi\rangle$.
Ans. 0 ; Expl. : $\psi(x) = x(l-x)$ is even w.r.t $x = \frac{l}{2}$ and $\psi_2(x) = \sqrt{\frac{2}{l}}\sin\frac{2\pi x}{l}$ has a node at $x = \frac{l}{2}$

$\psi(x)$ (even) PB $\psi_{n=2}$(odd)





$\therefore \langle 2|\psi\rangle = \int \psi_2(x)\psi(x)dx = 0$

3. (i) Using annihilation operator, $\hat{a} = \frac{1}{\sqrt{2}x_0}(\hat{x} + \frac{i}{\mu\omega}\hat{p})$,obtain the quantity $\langle m|x|n\rangle$for two harmonic oscillator eigenstates $|m\rangle$ and$|n\rangle$.
Expl.: $\langle m|x|n\rangle = \frac{x_0}{\sqrt{2}}\langle m|a+a^\dagger|n\rangle = \frac{x_0}{\sqrt{2}}(\langle m|a|n\rangle + \langle m|a^\dagger|n\rangle) = \frac{x_0}{\sqrt{2}}(\sqrt{n}\langle m|n-1\rangle + \sqrt{n+1}\langle m|n+1\rangle) = \frac{x_0}{\sqrt{2}}(\sqrt{n}\delta_{m,n-1} + \sqrt{m}\delta_{m,n+1})$.

(ii) Normalise the wavefunction for the lowest energy state of a harmonic oscillator $\left[\text{use } \int_{-\infty}^{\infty} e^{-z^2}dz = \sqrt{\pi}\right]$:.
Ans. Say, normalization constant $= N_0$: $\langle u_0|u_0\rangle = 1$
or, $N_0^2 \int_{-\infty}^{\infty}\left(e^{-\frac{\xi^2}{2}}\right)^2 dx = 1$; or, $N_0^2 x_0 \int_{-\infty}^{\infty} e^{-\xi^2}d\xi = 1$; or, $N_0^2 x_0\sqrt{\pi} = 1 \implies N_0^2 = \frac{1}{\sqrt{\pi}x_0}$, $\therefore N_0 = \frac{1}{(\pi x_0^2)^{\frac{1}{4}}}$

(iii) The equilibrium bond distance of the hydrogen molecule is 74 pm. Obtain an expression for the probability that in its lowest energy state, it has a bond distance between $148 - 150$ pm is nearly (N = normalisation constant for the state in $\text{pm}^{-\frac{1}{2}}$; $x_0 = \sqrt{\frac{\hbar}{\mu\omega}}$)
Ans. The given distance is twice the bond length, where the gaussian function for the lowest energy state varies smoothly over negligible values.
$\therefore$required prob. $= \int_{148-74}^{150-74} N_0^2 e^{-\frac{x^2}{x_0^2}}dx \approx (2\text{pm})\,.\,\left(N_0^2\text{pm}^{-1}\right).e^{-\frac{5625}{x_0^2}}$

(iv) Obtain the classical turning points for a harmonic oscillator in its 4th excited state ($n = 4$).
Ans. turning points: $x^2_{\text{t.pt.}_n} = \frac{2E_n}{k}$;
$x_{\text{t.pt.}} = \pm\sqrt{\frac{2E_n}{k}}$; $\xi_{\text{t.pt.}} = \pm\frac{\sqrt{\frac{2E_n}{k}}}{x_0} = \pm\frac{\sqrt{2\left(n+\frac{1}{2}\right)\frac{\hbar\omega}{k}}}{x_0} = \pm\sqrt{2n+1}.\frac{1}{x_0}\sqrt{\frac{\hbar\omega}{k}} = \pm\sqrt{2n+1}\overset{n=4}{=}\pm 3 \implies x_{\text{t.pt.}} = \pm 3x_0$

4. Two Hermitian operators $\hat{A}$ and $\hat{B}$ do not commute. Let $(\hat{A}+i\lambda\hat{B})|\psi\rangle = |\phi\rangle$, and $g = \langle\phi|\phi\rangle$

Show: (i) $g = \langle A^2\rangle + \lambda^2\langle B^2\rangle + \lambda\langle i[\hat{A},\hat{B}]\rangle$; (ii) $i[\hat{A},\hat{B}]$ is Hermitian.

(iii) the minimum value g (with respect to λ), $g_{\mathbf{min}} = \langle\widehat{A^2}\rangle - \frac{1}{4}\frac{\langle i[\hat{A},\hat{B}]\rangle^2}{\langle\widehat{B^2}\rangle}$

(iv) $\langle\hat{A}^2\rangle\langle\hat{B}^2\rangle \geq \frac{1}{4}\langle i[\hat{A},\hat{B}]\rangle^2$; (v) if $\hat{A} = \hat{p_x} - \overline{p_x}$ (mean deviation) and $\hat{B} = \hat{x} - \overline{x}$, then $\triangle p_x \triangle x \geq \frac{\hbar}{2}$, where $\triangle p_x = \sqrt{(\hat{p}-\overline{p})^2}$; and $\triangle x = \sqrt{(\hat{x}-\overline{x})^2}$

Ans. (i) $g = \langle\psi|(\hat{A}-i\lambda\hat{B})(\hat{A}+i\lambda\hat{B})|\psi\rangle = \langle A^2\rangle + \lambda^2\langle B^2\rangle + i\lambda\langle\psi|\hat{A}\hat{B}-\hat{B}\hat{A}|\psi\rangle = \langle A^2\rangle + \lambda^2\langle B^2\rangle + \lambda\langle i[\hat{A},\hat{B}]\rangle$

(ii) $\langle\psi_1|i[A,B]|\psi_2\rangle = \langle\psi_1|i(\hat{A}\hat{B}-\hat{B}\hat{A})|\psi_2\rangle = i\langle\psi_1|\hat{A}\hat{B}|\psi_2\rangle - i\langle\psi_1|\hat{B}\hat{A}|\psi_2\rangle$
$= i\langle\hat{A}\psi_1|\hat{B}|\psi_2\rangle - i\langle\hat{B}\psi_1|\hat{A}|\psi_2\rangle = i\langle\hat{B}\hat{A}\psi_1|\psi_2\rangle - i\langle\hat{A}\hat{B}\psi_1|\psi_2\rangle = \langle -i\hat{B}\hat{A}\psi_1|\psi_2\rangle + \langle i\hat{A}\hat{B}\psi_1|\psi_2\rangle$
$= \langle(i[A,B])\,\psi_1|\psi_2\rangle \quad \therefore i[\hat{A},\hat{B}]$ is Hermitian.

(iii) The function $g(\lambda) = a + b\lambda + c\lambda^2$ has a minimum when $\frac{\partial g}{\partial\lambda} = 0 \implies \lambda = -\frac{b}{2c}$ and $g_{\min} = a - \frac{b^2}{2c} + \frac{b^2}{4c} = a - \frac{b^2}{4c}$
$\therefore g_{\min} = \langle\hat{A}^2\rangle - \frac{1}{4}\frac{\langle i[\hat{A},\hat{B}]\rangle^2}{\langle\hat{B}^2\rangle}$

(iv) $g_{\min}$ is never negative. Even at its minimum value $g_{\min} \geq 0$
$\implies \langle\hat{A}^2\rangle\langle\hat{B}^2\rangle \geq \frac{1}{4}\langle i[\hat{A},\hat{B}]\rangle^2$

(v) for $\hat{A} = \hat{p_x} - \overline{p_x}$ and $\hat{B} = \hat{x} - \overline{x}$, $\quad i[\hat{A},\hat{B}] = i(-i\hbar) = +\hbar$
$\implies \langle(\hat{p}-\overline{p})^2\rangle\langle(\hat{x}-\overline{x})^2\rangle \geq \frac{\hbar^2}{4}$[using $[\hat{p}_x, \hat{x}] = -i\hbar\hat{1}$
$\implies \triangle p_x \triangle x \geq \frac{\hbar}{2}$. This is the famous Uncertainty principle. [Note : Here, Δ is the standard deviation : uncertainty in measurement]